

Autoimmunity Links Vinculin to the Pathophysiology of Chronic Functional Bowel Changes Following *Campylobacter jejuni* Infection in a Rat Model

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Abstract

Background Acute gastroenteritis can precipitate irritable bowel syndrome (IBS) in humans. Cytotoxic distending toxin is common to all pathogens causing gastroenteritis. Its active subunit, CdtB, is associated with post-infectious bowel changes in a rat model of *Campylobacter jejuni* infection, including small intestinal bacterial overgrowth (SIBO).

Aim To evaluate the role of host antibodies to CdtB in contributing to post-infectious functional sequelae in this rat model.

Methods Ileal tissues from non-IBS human subjects, *C. jejuni*-infected and control rats were immunostained with antibodies to CdtB, c-Kit, S-100, PGP 9.5 and vinculin. Cytosolic and membrane proteins from mouse enteric neuronal cell lysates were immunoprecipitated with

anti-CdtB and analyzed by mass spectrometry. ELISAs were performed on rat cardiac serum using CdtB or vinculin as antigens.

Results Anti-CdtB antibodies bound to a cytosolic protein in interstitial cells of Cajal (ICC) and myenteric ganglia in *C. jejuni*-infected and naïve rats and human subjects. Mass spectrometry identified vinculin, confirmed by co-localization and ELISAs. Anti-CdtB antibodies were higher in *C. jejuni*-infected rats (1.27 ± 0.15) than controls (1.76 ± 0.12) ($P < 0.05$), and rats that developed SIBO (2.01 ± 0.18) vs. rats that did not (1.44 ± 0.11) ($P = 0.019$). Vinculin expression levels were reduced in *C. jejuni*-infected rats (0.058 ± 0.053) versus controls (0.087 ± 0.023) ($P = 0.0001$), with greater reductions in rats with two *C. jejuni* infections ($P = 0.0001$) and rats that developed SIBO ($P = 0.001$).

Conclusions Host anti-CdtB antibodies cross-react with vinculin in ICC and myenteric ganglia, required for normal gut motility. Circulating antibody levels and loss of vinculin expression correlate with number of *C. jejuni* exposures and SIBO, suggesting that effects on vinculin are

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important in the effects of *C. jejuni* infection on the host gut.

Keywords Irritable bowel syndrome · Cytolethal distending toxin · Vinculin · Small intestinal bacterial overgrowth

Introduction

Irritable bowel syndrome (IBS) is one of the most common human disorders, affecting 10–20 % of adults worldwide [1, 2] and accounting for 50 % of all gastrointestinal office visits [3, 4]. In a time of economic burdens, IBS accounts for 30 % of all costs in gastroenterology [3–6]. However, progress in the diagnosis and treatment of IBS has been hampered by its diversity of symptoms, which include abdominal pain, bloating and altered bowel function, by its differing presentations, which include diarrhea-predominant (D-IBS), constipation-predominant (C-IBS) and alternating forms [7], and by the lack of specific laboratory diagnostic tests. As a result, the diagnosis of IBS has been a clinical one based on the Rome Criteria [7–12], and IBS is frequently a “diagnosis of exclusion” reached only after lengthy testing which is both costly and distressing to the patient [13]. Although the pathogenesis of IBS remains unclear, recent hypotheses from two distinct research areas suggest that aberrations of the gut flora may be involved. The first is that small intestinal bacterial overgrowth (SIBO) contributes to the symptoms of IBS. Although there has been considerable controversy surrounding the potential contribution of SIBO to IBS, stemming partly from the lack of an accepted “gold standard” for SIBO [14] as well as from debates surrounding breath testing techniques [14, 15], this hypothesis is supported by breath test studies indicating that SIBO occurs in up to 84 % of IBS subjects [16–20], by culture studies demonstrating bacterial overgrowth in the proximal small bowel in IBS [21, 22], and by phase III trials reporting the success of antibiotics in treating IBS [23]. The second is that “post-infectious” IBS (PI-IBS) can develop after an episode of acute gastroenteritis, commonly caused by *Campylobacter jejuni* [24], *Salmonella* [25], *Escherichia coli* [26] and *Shigella* [27]. Two meta-analyses suggest that up to 10 % of gastroenteritis sufferers subsequently develop IBS [28, 29].

To determine the mechanisms underlying the development of IBS, we developed an animal model using *C. jejuni*, which is a common cause of acute gastroenteritis in the USA [30], as the infective agent. Results from this model demonstrate that acute gastroenteritis can precipitate the development of bacterial overgrowth [31]. Animals

infected with *C. jejuni* developed functional and structural changes long after clearance of the initial infection, including altered stool form, significant increases in rectal and left colon intraepithelial lymphocytes, lower body weights and SIBO [31]. Further, the presence of SIBO was found to be predictive of altered stool form in these animals [31]. In human IBS subjects, the development of SIBO is associated with a reduced frequency of phase III of interdigestive motor activity [32, 33], which is likely due to the loss of interstitial cells of Cajal (ICC) and myenteric nerves [32, 34]. Significantly, this rat model also exhibited significantly reduced numbers of ICC in the deep muscular plexus [35], and reduction in ICC numbers was correlated with the development of SIBO [35]. Visceral pain has not been examined in this model to date, although future studies will include determining visceral sensitivity in these animals. The results obtained to date provide a potential link between acute gastroenteritis and the development of these phenotypes in this rat model.

Cytolethal distending toxin (Cdt) is common to most of the pathogens which cause acute gastroenteritis and IBS in humans. Cdt is a heterotrimeric toxin (CdtABC) with cytolethal distending toxin B (CdtB) being the active subunit. Recently, we showed that animals infected with a mutant *C. jejuni* strain lacking CdtB (due to an insertional deletion) exhibited significantly reduced bowel disturbances compared to those infected with wild-type *C. jejuni* [36], which suggested that CdtB might play an important role in the pathophysiology of IBS. In this study, we evaluate the role of CdtB and host antibodies produced against CdtB in the development of the gut phenotypes seen in this rat model.

Materials and Methods

Animal Tissue and Serum Samples

Ileal tissue sections were obtained from rats that were gavaged with 10^8 cfu/mL of *C. jejuni* 81-176 and euthanized on day 2 post-gavage and from uninfected controls [31]. Jejunal tissue samples and cardiac sera were obtained from three groups of rats: Group 1: uninfected controls ($n = 20$); Group 2 (single infection): rats gavaged with PBS as juveniles (i.e., 24 days of age) and then with 10^8 cfu/mL *C. jejuni* 81-176 2 months later as adults ($n = 50$); and Group 3 (double infection): rats gavaged with 10^8 cfu/mL of *C. jejuni* 81-176 first as juveniles and again as adults ($n = 50$) [37]. Rats were euthanized by CO₂ asphyxiation 3 months following clearance of *C. jejuni* from stool after the adult infection. Immediately following CO₂ asphyxiation, cardiac blood was collected to separate serum for analysis by ELISA, and tissue sections and luminal contents from duodenum, jejunum and ileum were harvested

[37]. SIBO levels were determined by qPCR [37] and defined as >2 standard deviation above the average of controls in accordance with previous studies [31]. All animal work was approved by the Cedars-Sinai Institutional Animal Care Utilization Committee (IACUC).

Human Tissue Samples

Human ileal samples were obtained from subjects who underwent hemicolectomy for colon cancer, excluding those who underwent surgery for any inflammatory bowel disease (IBD). These deidentified human tissue samples were obtained under a protocol approved by the Cedars-Sinai Institutional Review Board (IRB).

Immunohistochemical and Immunofluorescent Microscopy

Both immunohistochemical and immunofluorescent staining were performed to compare anti-CdtB (provided by Dr. P. Guerry, see Supplemental Information) to markers for ICC (c-Kit), neurons (S-100) and ganglia (PGP 9.5), as well as vinculin. Briefly, slides were deparaffinized and rehydrated through a graded ethanol series, and heat-treated antigen retrieval was performed using 10 mM sodium citrate buffer, pH 6.0. Sections were blocked with 10 % chicken serum and 10 % donkey serum in 2 % BSA/PBS for 1 h at room temperature, and then incubated with anti-CdtB (1:500 dilution) either alone or with anti-S-100 (1:100, Pierce Biotechnology, Rockford, IL), anti-PGP 9.5 (1:10, Pierce Biotechnology, Rockford, IL), anti-c-Kit (1:33, R&D Systems, Minneapolis, MN) or anti-vinculin (1:10, Santa Cruz Biotechnology, Santa Cruz, CA) overnight at 4 °C. For immunohistochemistry, secondary antibody incubations and counterstaining were performed as described previously [35]. For immunofluorescence, sections were incubated for 2 h with goat anti-rabbit Alexa Fluor-488, either anti-goat or anti-mouse Alexa Fluor-568, and DAPI, and then washed and mounted using Prolong Gold (Invitrogen, Grand Island, NY).

Immunoprecipitation and Identification of the Enteric Neuronal Protein

Cell Lysis

Murine enteric undifferentiated neuronal cells [2.35×10^8 cells, provided by Dr. Shanthi Srinivasan (Emory University, Atlanta, GA)] were lysed using the *DUALXtract* Membrane Protein Extraction Kit (Dualsystems Biotech) according to the manufacturers' recommendations. Briefly, 3.5 ml of ice-cold cell permeabilization buffer including Halt protease and phosphatase inhibitor cocktail (Thermo Scientific) was added

to the cell pellet, and the mixture was vortexed and incubated for 10 min at 4 °C with continuous rocking. The permeabilized cells were centrifuged at $16,000 \times g$ for 15 min at 4 °C, and the cytosolic fraction (supernatant) was collected. The membrane fraction was prepared by adding 2.35 ml ice-cold membrane protein extraction buffer (containing Halt protease and phosphatase inhibitor cocktail) to the remaining pellet, which was mixed by pipetting and incubated for 30 min at 4 °C with shaking. The mixture was centrifuged at $16,000 \times g$ for 15 min at 4 °C, and the membrane fraction (supernatant) was saved. The cytosolic and membrane fractions of the lysate were then precleared by passing each through a column containing 235 μ l packed Protein A agarose beads (Calbiochem-EMD Millipore) equilibrated with either cell permeabilization or membrane protein extraction buffer, respectively.

Antibody Crosslinking

Two hundred and twenty-five microliters of Protein A agarose slurry was centrifuged at 6,000 rpm, and the packed beads (approximately 150 μ l) were washed three times with 0.2 M triethanolamine, pH 9.0. To the packed beads, 19 μ l of CdtB antiserum (wAb) (approximately 470 mg protein) was added, and the volume of the mixture was adjusted to 10 ml using 0.2 μ M triethanolamine buffer. Two hundred and eighty milligrams of dimethyl pimelimidate dihydrochloride (Sigma) was added, and the mixture was rotated for 1 h at room temperature. The coated beads were washed twice with triethanolamine, rotated in 10 ml of fresh buffer for 2 h at room temperature, washed 3 times with 10 ml PBS, resuspended in PBS in a total volume of 450 μ l and stored at 4 °C.

Immunoprecipitation, SDS-PAGE and Mass Spectrometry

For immunoprecipitations, 35 μ l of packed beads coated with antibodies to CdtB were washed 3 times with either cell permeabilization buffer or membrane protein extraction buffer, mixed with the pre-cleared cytosolic or membrane fractions, respectively, and rotated for 4 h at room temperature. The beads were then centrifuged at 6,000 rpm at 4 °C, washed three times in permeabilization or membrane protein extraction buffer, respectively, and washed once with PBS. The immunoprecipitates were added to SDS protein denaturing buffer (Thermoscientific) and heated at 100 °C for 5 min. The proteins were then separated on a 12 % SDS-polyacrylamide gel, stained using Sypro Ruby protein gel stain (Invitrogen) and visualized on a UV light box (see Supplemental Figure 4). The bands of interest were excised and submitted to the UCLA W. M. Keck Proteomic Center for trypsinization and NanoLC-MS/MS-Protein identification.

Enzyme-Linked Immunosorbent Assay (ELISA) Analysis of Rat Sera

ELISAs were conducted using a recombinant CdtB (provided by Dr. Kenneth Bradley, MIMG Department, UCLA) or vinculin (Novoprotein, Short Hills, NJ) as the antigens. Antigens (1.2 µg/ml) were immobilized overnight at 4 °C onto high-binding 96-well plates (Grenier Bio-One, Monroe, NC) in Borate Buffered Saline (Medicago, Uppsala, Sweden) at a pH of 8.2. Individual plates were coated with either recombinant CdtB or vinculin, and each plate contained both coated and uncoated wells, in order to determine and control for non-specific binding of sera. Wells were blocked with 3 % BSA in PBS for 1 h at room temperature. Rat sera from all 3 groups of rats [37] were diluted 1:16 dilution and added in duplicate to both coated and uncoated wells for 1 h at room temperature. Antibodies to CdtB (wAb) and vinculin were used as positive controls. HRP-conjugated secondary antibodies (Jackson ImmunoResearch, West Grove, PA) were then added to the wells and incubated for 1 h. Each step was followed by a series of plate washes using 0.05 % PBS-T. Finally, a TMB substrate solution (Pierce, Rockford, IL) was used for visualization and immediately read on a BioTek Synergy HT plate reader. Levels of antibodies to CdtB and vinculin were compared based on the number of *C. jejuni* infections and the presence or absence of SIBO.

Evaluating Vinculin Gene Expression Levels in Jejunal Tissues

Total RNA was extracted from jejunal tissue samples from all 3 groups of rats [37] using AllPrep DNA/RNA Mini Kits (Qia-gen) according to manufacturer's instructions. Complementary DNA (cDNA) was synthesized from 1 µg total RNA using the iScript cDNA Synthesis Kit (Bio-Rad), again according to manufacturer's instructions. Quantitative real-time PCR (q-RT-PCR) was then performed to quantify the amounts of the different transcripts present, using 1 µl of reverse transcription reaction in 25 µl q-RT-PCR. Q-RT-PCR was performed on a MyiQ Thermal Cycler (Bio-Rad) with iQ SYBR green supermix (Bio-Rad). The primers for vinculin were 5'-GCC AAGCAGTGCACAGATAA-3' (For) and 5'-TCTTTCTG ACCCAGCGCAGA-3' (Rev), and the primers for β-actin were 5'-GGAGATTACTGCCCTGGCTCCTA-3' (For) and 5'-GACTCATCGTACTCCTGCTTGCTG-3' (Rev). Vinculin expression levels were calculated by normalizing the cycle numbers to those of β-actin and compared based on the number of *C. jejuni* infections and the presence or absence of SIBO.

Statistical Analysis

Comparisons of anti-CdtB and anti-vinculin levels between groups, and of tissue vinculin gene expression levels in

C. jejuni-infected and control rats, were done by the non-parametric Mann–Whitney *U* test. Correlation curves between bacteria counts and levels of antibodies to CdtB were compared by Pearson rank correlation. When comparing tissue vinculin levels between control, single-infected and double-infected animals, a Kruskal–Wallis test was used. The prevalence of SIBO between single-infected and double-infected animals was compared by Chi square test. Significance was set at $P < 0.05$, and data are expressed as mean ± SEM.

Results

Antibodies to CdtB Co-localize with Markers for ICC and Enteric Neurons and Ganglia

To determine the role of the CdtB toxin in the development of the gut phenotypes seen in this rat model, we used a validated antibody generated to near-full-length *C. jejuni* CdtB (see Supplementary Information). Sections of rat ileum were examined 2 days after gavage with live *C. jejuni* 81-176. Extensive CdtB staining was observed in the mucosal surface and crypts (Supplemental Figure 2). Deep tissue components most identified were ICC, myenteric ganglia and other neural structures (Fig. 1a and Supplemental Figure 2b). However, the same pattern of staining was seen in ileal sections of rats that were never exposed to *C. jejuni* (Fig. 1b and Supplemental Figure 2d) and in sections of full-thickness ileal tissue from human colon cancer patients who did not have IBS or IBD (Fig. 2 and Supplemental Figure 3). Co-localization confirmed that CdtB antibodies bind to ICC as well as enteric neurons, both in *C. jejuni*-infected rats and in control rats that had not been infected with *C. jejuni* (Fig. 1c, d), as well as in ileal sections from human subjects who did not have IBS or IBD (Fig. 2). This pattern of staining, which was localized to the cytosolic component in both ICC and neurons, suggested that antibodies to the CdtB toxin can also bind to an enteric neuronal protein.

Antibodies to CdtB Cross-React with the Enteric Neural Protein Vinculin

To identify the enteric neuronal protein with which antibodies to CdtB cross-react, murine enteric undifferentiated neuronal cells were lysed and probed with antibodies to CdtB on Western blots. A 117-kDa band was identified which was eliminated when the antibodies were blocked with the whole CdtB protein. Fractionation of these lysates revealed that the 117-kDa protein was located in the cytosolic fraction (see Supplementary Information). The lysates were then immunoprecipitated with the anti-CdtB antibody,

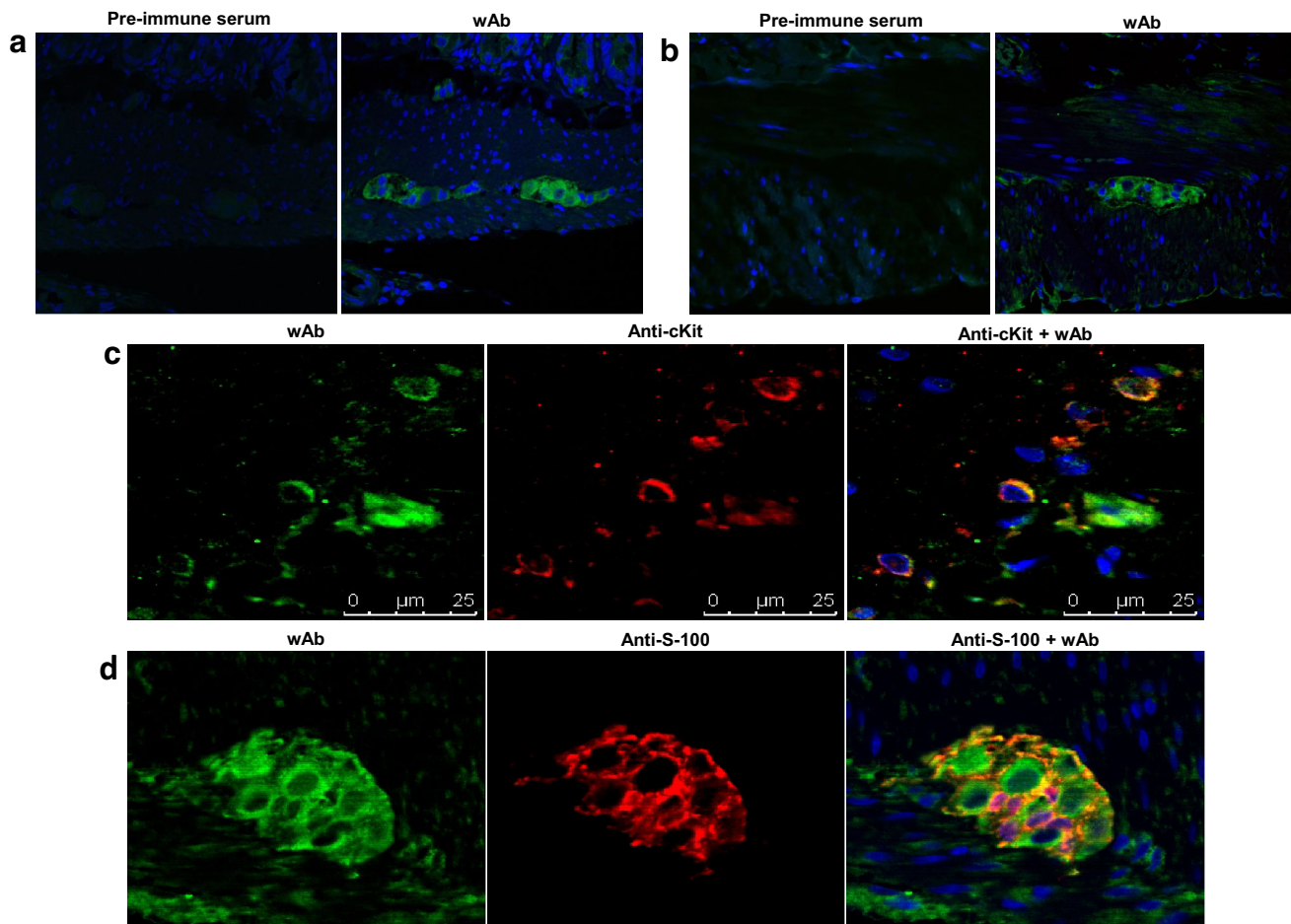


Fig. 1 Antibodies to CdtB stain endogenous neural elements of the gut in the rat ileum. **a, b** Immunofluorescent staining of rat ileal tissue sections with CdtB antiserum or pre-immune serum in *C. jejuni*-

infected rats 2 days post-gavage (**a**) or control rats (**b**). **c, d** Antisera to CdtB co-localize with markers for ICC (c-kit) (**c**) and for enteric neurons (S-100) (**d**)

and the 117-kDa band from a gel of the immunoprecipitates (Supplemental Figure 4) was excised and sent for NanoLC-MS/MS-Protein identification. The results identified the protein candidate in this band as vinculin. Subsequent staining confirmed that antibodies to vinculin and CdtB co-localize neural elements in sections of human ileal tissue (Fig. 3a), further supporting vinculin as the enteric neural protein with which antibodies to CdtB cross-react. This was also supported by an ELISA testing antibodies to CdtB and vinculin against vinculin as the antigen (Fig. 3b).

Antibodies to CdtB Are Elevated in *C. jejuni*-Infected Rats and Correlate with Levels of SIBO

To determine the role of circulating antibodies to CdtB in the gut phenotypes seen in this rat model, an ELISA was developed using recombinant *C. jejuni* CdtB or vinculin as antigens. Sera from control rats ($n = 10$), rats exposed to *C. jejuni* once ($n = 20$) and rats exposed to *C. jejuni* twice (2 months apart,

$n = 20$) [37] were tested using this ELISA, and the results were correlated with the presence and level of SIBO as determined by qPCR. For circulating levels of antibodies to CdtB, the optical densities (ODs) were 1.27 ± 0.15 in the control rats ($n = 10$) and 1.76 ± 0.12 in all rats exposed to *C. jejuni* ($n = 40$) ($P < 0.05$) (Fig. 4a), with highest levels in rats that were exposed to *C. jejuni* twice and which had high bacterial counts in the small bowel (Fig. 4a). For rats exposed to *C. jejuni*, those that developed SIBO had higher ODs (2.01 ± 0.18 , $n = 23$) than those that did not develop SIBO (1.44 ± 0.11 , $n = 17$) ($P = 0.019$), irrespective of number of exposures to *C. jejuni* (Fig. 4b). Similar findings were obtained when vinculin was used as the ELISA substrate (Fig. 4c).

Vinculin Gene Expression in Rats Based on Number of Exposures to *C. jejuni* and SIBO

The levels of vinculin gene expression in the wall of the jejunum were determined by q-RTPCR, by normalizing the

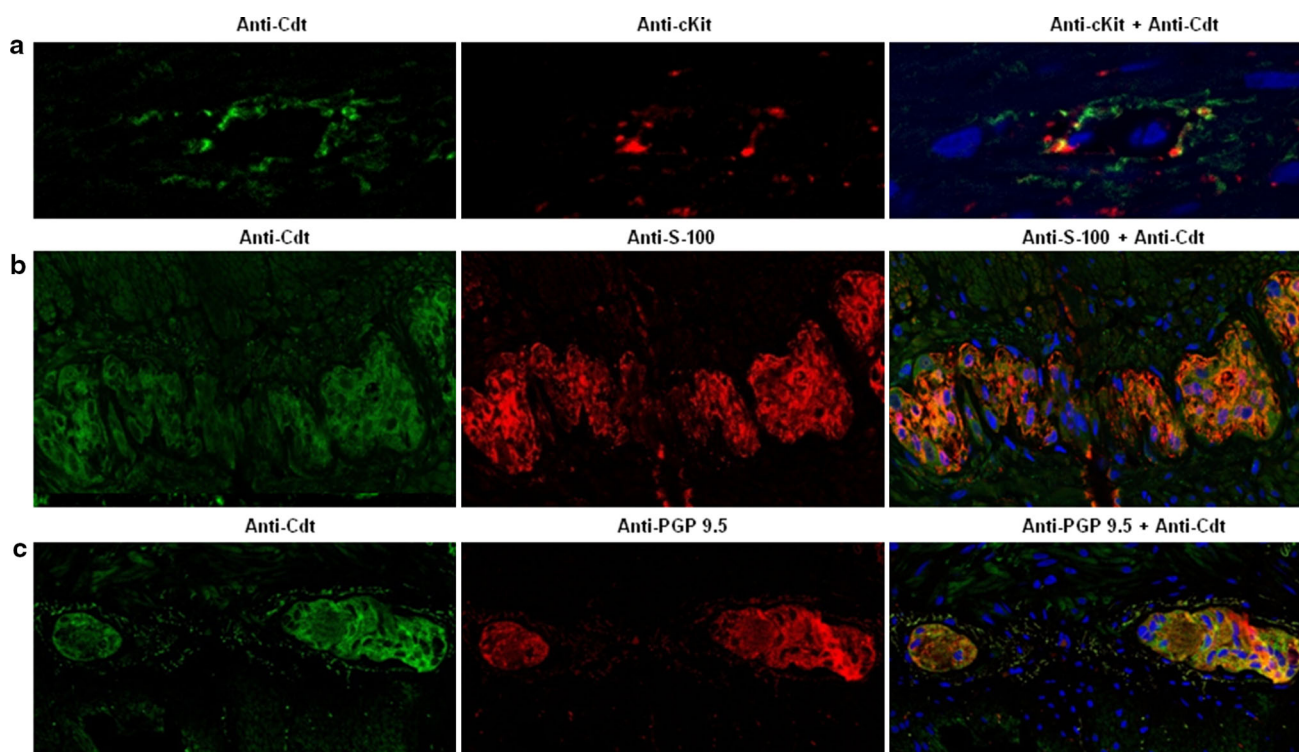


Fig. 2 Antibodies to CdtB stain endogenous neural elements of the gut in the human ileum. Antisera to CdtB co-localize with markers for ICC (c-kit) (a), enteric neurons (S-100) (b) and ganglia (PGP 9.5) (c) in ileal tissues from human subjects who did not have IBS

cycle numbers to those of β -actin in control rats ($n = 20$) and rats exposed to *C. jejuni* once ($n = 50$) or twice ($n = 49$). Vinculin gene expression in the jejunum was reduced in rats exposed to *C. jejuni* (0.058 ± 0.053) compared to control rats (0.087 ± 0.023) ($P = 0.0001$) (Fig. 5). There was a greater reduction of vinculin in rats that had two exposures (0.025 ± 0.012) to *C. jejuni* compared to a single exposure (0.090 ± 0.058) ($P = 0.0001$) (Fig. 6a). SIBO during the post-infectious phase was defined to be present when bacterial counts for a single segment of small bowel in the *Campylobacter*-infected rats exceeded two standard deviations above the average of controls as described above. Vinculin gene expression levels were lower in rats that developed SIBO after *C. jejuni* infection (0.052 ± 0.054 , $n = 36$) than in those that did not (0.061 ± 0.053 , $n = 63$) ($P = 0.001$) (Fig. 6b).

Discussion

Our results demonstrate that autoimmunity may play an important role in the host response to infection with *C. jejuni*. Infection with *C. jejuni* is a leading cause of gastroenteritis, which in turn can lead to the development of post-infectious IBS (PI-IBS) in humans [28, 29]. Antibodies produced by the host in response to the *C. jejuni* CdtB subunit cross-react with

elements of the enteric nervous system, specifically ICC and myenteric ganglia. The ICC and myenteric ganglia are known to be required for normal intestinal motility, including phase III of interdigestive motor activity [38–46], and the absence or reduced frequency of phase III contractions induces SIBO both in animal models [34] and in human subjects, including those with IBS [32]. The results from this animal model of *C. jejuni* infection suggest that antibodies to CdtB produced following infection with *C. jejuni* bind to vinculin in the ICC and myenteric ganglia, likely through molecular mimicry between vinculin and CdtB. We hypothesize that this may affect intestinal motor activity and lead over time to the development of SIBO. Consistent with this hypothesis, we found that higher levels of circulating antibodies to CdtB correlated with levels of SIBO in our rat model. We also found that vinculin expression levels in the intestinal wall are reduced following exposure to *C. jejuni*. This reduction is dependent upon the number of exposures to *C. jejuni*, with greater reduction in rats that have been exposed to *C. jejuni* twice. Lastly, vinculin expression levels were also reduced in rats that developed SIBO. These findings suggest that loss of vinculin in the neuromuscular apparatus of the gut may contribute to the gut phenotypes in this rat model following *C. jejuni* infection.

In the last decade, it has become very clear that acute gastroenteritis is a cause of post-infectious IBS. Two meta-

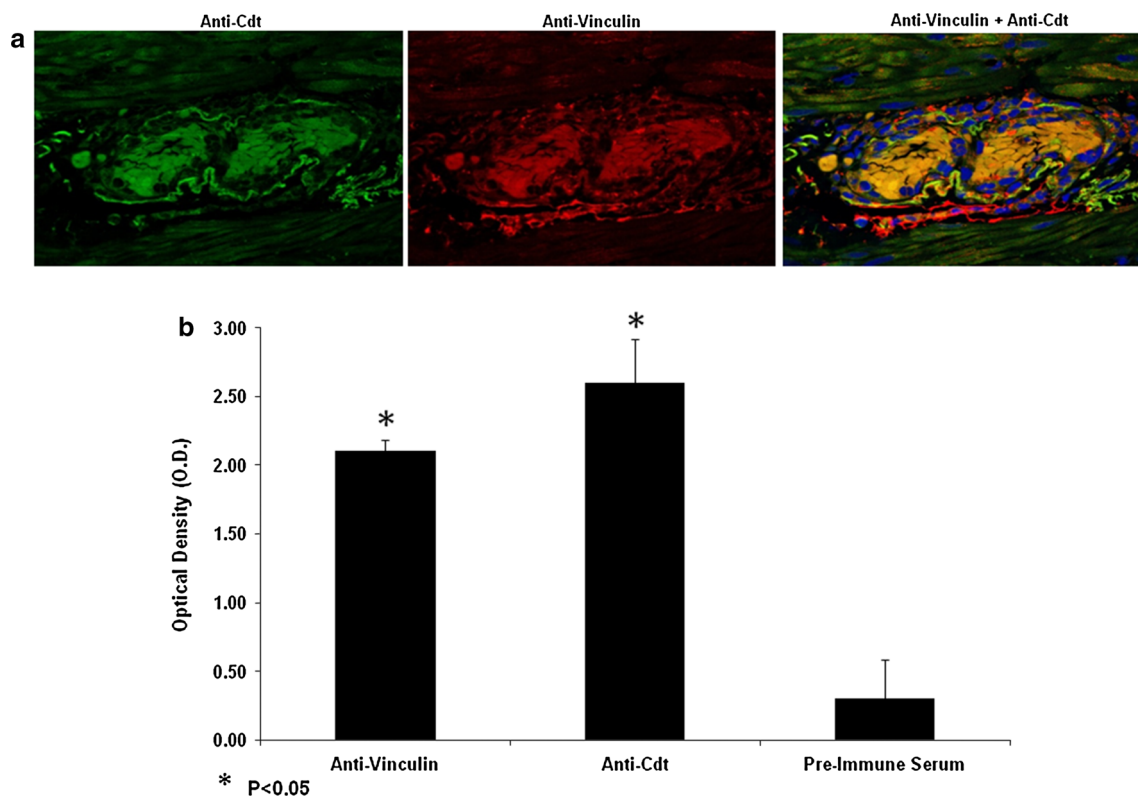


Fig. 3 Vinculin expression in human ileum and detection in a novel ELISA. **a** Antibodies to vinculin and CdtB co-localize in ileal tissues from human subjects who did not have IBS. **b** Binding of anti-

vinculin, anti-CdtB and pre-immune serum to recombinant vinculin protein on ELISA (as a test of ELISA specificity). * $P < 0.05$, error bars denote standard error of the mean (SEM)

analyses indicate that the incidence of IBS after an outbreak of bacterial gastroenteritis is approximately 10 % [28, 29]. Two important outbreaks have been mostly studied: the Walkerton outbreak in Canada [47] and the *Salmonella* outbreak in Spain [25]. Prior to these studies, investigators had suggested that PI-IBS was either a separate entity or a small subset of the total IBS population. However, a recent model using military and the known prospective data on PI-IBS, combined with CDC data on the incidence of gastroenteritis in the USA, suggests that more than 9 % of the entire US population could have IBS precipitated by gastroenteritis [48]. Although modeling can be problematic, at a minimum this suggests that gastroenteritis could be responsible for a large portion of IBS cases and may in fact be the major cause.

Numerous animal models have been created to study PI-IBS [49]. However, some of the more prominently published models have used pathogens that are uncommon in IBS. In contrast, *C. jejuni* is the most common cause of bacterial gastroenteritis in the USA [30] and thus is likely the most significant contributor to the overall incidence of PI-IBS in the USA. Studies over a decade ago began to suggest that SIBO might be a feature of IBS. Two recent culture-based studies have confirmed an excess of coliform

bacteria in the small intestine of IBS compared to healthy controls [21] and to subjects with other foregut disease [22]. Our rodent model using *C. jejuni* infection demonstrates increased rectal intraepithelial lymphocytes, weight loss and the development of altered stool form and SIBO [31, 35], similar to the findings in human subjects with PI-IBS, and also exhibits loss of ICC. Although we have not examined visceral pain in this model to date, future studies will include determining visceral sensitivity in these animals.

Multiple bacterial pathogens in addition to *C. jejuni* have been incriminated in the development of PI-IBS, suggesting either a common host response to infection or a common toxin. Cytolethal distending toxin (Cdt) is common to all pathogens causing acute gastroenteritis, which suggested that it might play a role in the development of IBS. In this study, we demonstrate that the effects of the active subunit CdtB on the host may result from the production of antibodies to CdtB that also recognize and bind to vinculin, a host cytosolic protein in the myenteric neurons and ICC. In our rat model of *C. jejuni* infection, levels of these antibodies correlate with loss of vinculin in the gut wall and with the development of SIBO. This cross-reaction also occurs in human gut wall tissues, which may

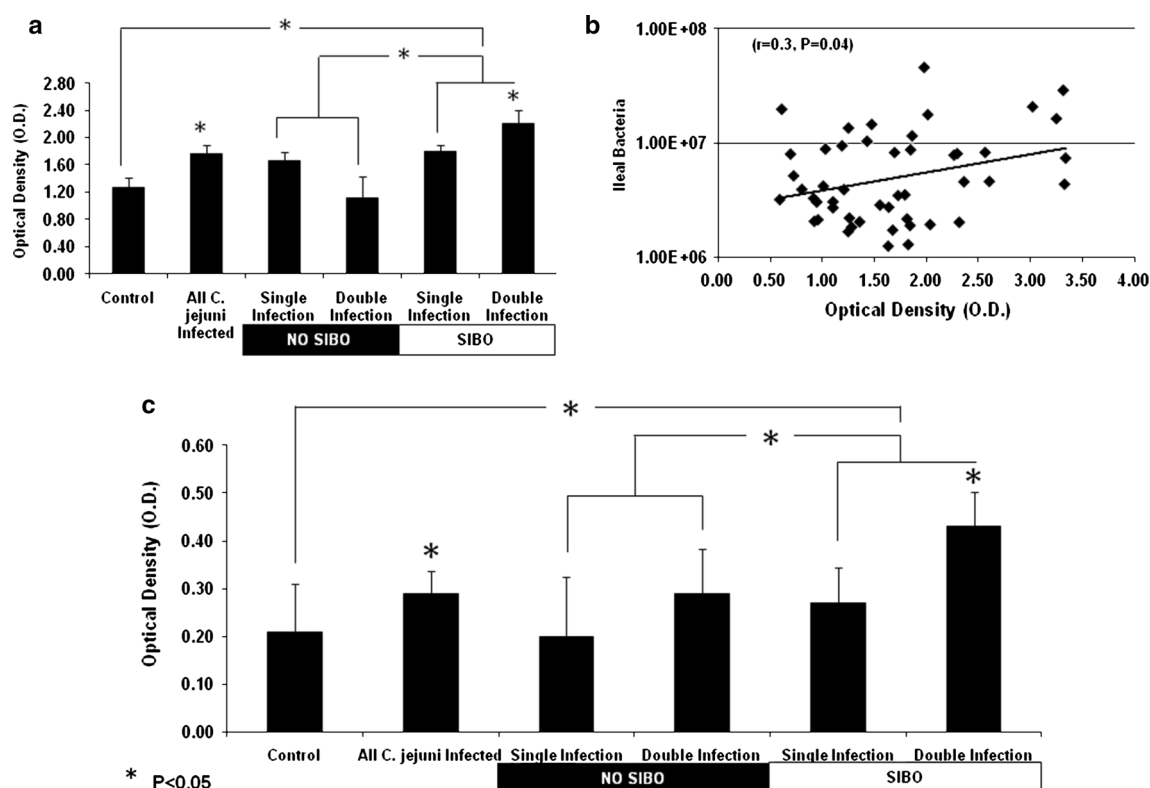


Fig. 4 Correlations between circulating levels of antibodies to CdtB, levels of SIBO and vinculin expression levels in *C. jejuni*-infected rats. **a, c** Levels of antibodies to CdtB in serum from control ($n = 10$), single-infected ($n = 20$) and double-infected ($n = 20$) rats with high or low bacterial counts in the small bowel, using recombinant *C.*

jejuni CdtB (**a**) or vinculin (**c**) as the ELISA substrate. **b** Correlation between levels of SIBO in the small bowel and titers of antibodies to CdtB in rat serum on ELISA ($r = 0.3, P = 0.04$). $^{\#}P < 0.001$, $^{\Delta}P < 0.0001$, error bars denote standard error of the mean (SEM)

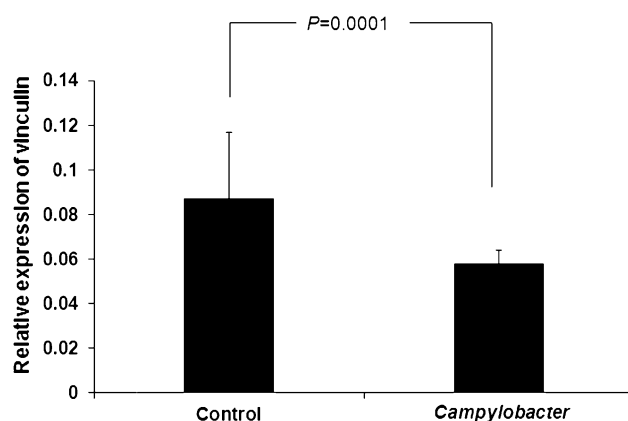


Fig. 5 Vinculin expression in rats inoculated with or without *C. jejuni*. Levels of vinculin expression in the jejunal wall in control rats ($n = 20$) versus those infected with *C. jejuni* once ($n = 50$) or twice ($n = 49$). Error bars denote standard error of the mean (SEM)

suggest that autoimmune mechanisms also contribute to the development of PI-IBS. In an independent study, a known anti-neuronal antibody was found to have a higher prevalence in IBS. However, this antibody was reactive to enteric nerve nuclei, and most of the healthy control

subjects also had these antibodies [50]. In contrast, vinculin is a cytoskeletal protein and important component of adherens junctions [51, 52] with functions in cell adhesion [53], neuronal cell motility and contractility [54], and structural integrity [55]. Therefore, it is plausible that impairment of its activity could contribute to the altered tissue histology and development of SIBO in our rat model, and might also contribute to reduced gut motility associated with IBS. An independent study recently found that Cdt from *Helicobacterium pullorum* targets vinculin in intestinal epithelial cells, triggering an atypical delocalization of vinculin from focal adhesions and formation of cortical large lamellipodia coupled with decreased cellular adherence [56]. These data support our findings that antibodies to *C. jejuni* CdtB bind to vinculin and that increased titers of antibodies to CdtB correlate with the degree of loss of vinculin in the intestinal wall in our rat model. Interestingly, molecular mimicry of the vinculin–talin interaction is the mechanism by which both the IpA toxin of *Shigella* [57] and the *Rickettsia* surface cell antigen 4 [58] achieve cell entry, activating vinculin and interacting with the actin cytoskeleton. Levels of antibodies to CdtB also correlated with levels of SIBO, which may be an indirect

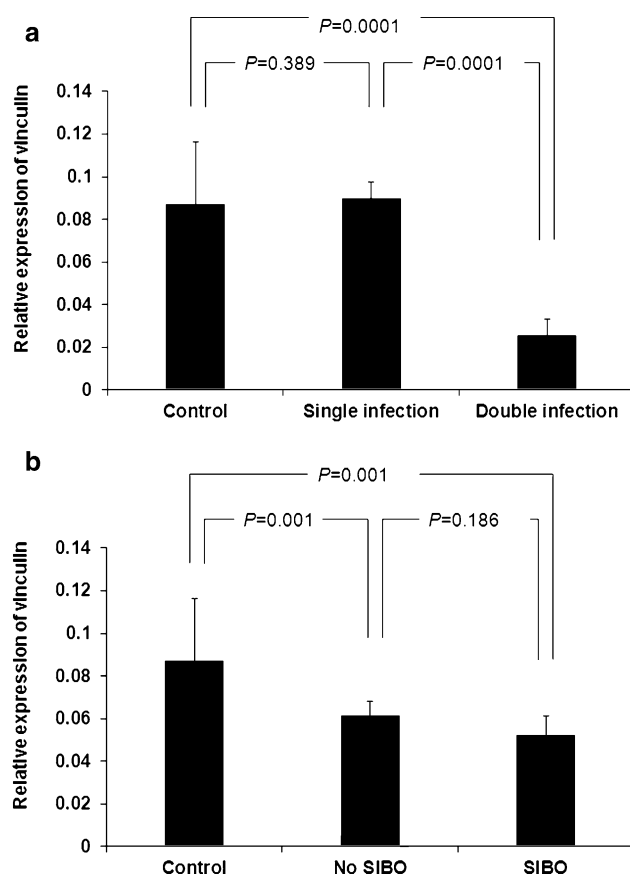


Fig. 6 Vinculin expression in rats based on the number of *C. jejuni* infections and levels of SIBO. Levels of vinculin expression in the jejunal wall in control ($n = 20$), single-infected ($n = 50$) and double-infected ($n = 49$) rats (a); and in control rats, *C. jejuni*-infected rats that did not develop SIBO ($n = 63$) and *C. jejuni*-infected rats that did develop SIBO ($n = 36$) (b). Comparisons between two groups were performed by the Mann–Whitney *U* test, and three-group comparisons were done by Kruskal–Wallis test. Error bars denote standard error of the mean (SEM)

measure of neuronal impairment in the small bowel, as we and others have shown that SIBO is a consequence of altered interdigestive motor complexes [32, 33] and that its eradication in IBS subjects results in improved motility [33]. The level of SIBO in turn appears to correlate with the level of altered bowel function, both in this rat model [31] and in human subjects [23]. Further, we recently found that levels of circulating antibodies to CdtB are increased in human subjects with IBS (Pimentel et al., submitted), which may suggest that this antibody may be important in humans as well. What remains to be demonstrated using this animal model is whether anti-CdtB antibodies are associated with vinculin activity and/or changes in gut motor function in vivo. This would further delineate a cause-and-effect relationship.

In conclusion, we have previously shown that the CdtB toxin is instrumental in the development of post-infectious gut changes in our rat model of *C. jejuni* infection [36]. In

this study, we demonstrate that the effects of CdtB on the host may result from the production of antibodies to CdtB that, through molecular mimicry, also bind to the host enteric nerve protein vinculin, and also demonstrate that circulating levels of these antibodies to CdtB are predictive of the development of SIBO in our rat model. Further, both the number of *C. jejuni* exposures and levels of SIBO correlate with loss of vinculin in the intestinal wall. This represents a breakthrough in understanding the roles of *C. jejuni* infection and the CdtB toxin in contributing to impairment of gut function.

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